

Functions and Relations

Week 4D

Overview

Domain and Range

Key Ideas - Finding the Domain and Range Graphically

- Finding the Domain and Range Algebraically

Function and Relations

Key Ideas - Piecemeal Functions

- Odd and Even Functions

Key Idea – Domain and Range

Definitions

The **domain** of a function $y = f(x)$ is the set of all **x values** for which $f(x)$ is defined. It represents the ‘**input**’ values of a function.

The **range** of a function $y = f(x)$ is the set of all **y values** obtained from the x values in the domain. It represents the ‘**output**’ values of a function.

***Note:** It is often easier to determine the range of a function by graphing it first.*

Key Idea – Finding Domain and Range Graphically

Domain: the domain is all x -values for which the graph exists.

Steps:

1. **Look left and right along the x -axis.**
2. **Find where the graph starts and ends horizontally.**
3. Write down the smallest and largest x -values shown.
4. Decide if the endpoints are:
 - **Included** (solid dot $\rightarrow$ use $[]$)
 - **Not included** (open dot $\rightarrow$ use $()$)
5. Write the answer in interval notation

Range: the range is all y -values for which the graph reaches.

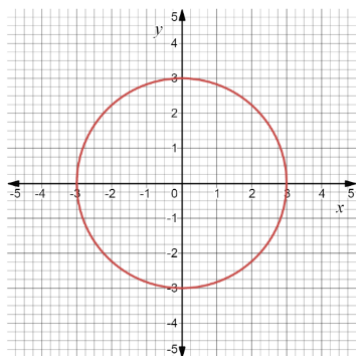
Steps:

1. **Look up and down along the y -axis.**
2. **Find the lowest and highest points on the graph.**
3. Write down the smallest and largest y -values shown.
4. Decide if the endpoints are:
 - **Included** (solid dot $\rightarrow$ use $[]$)
 - **Not included** (open dot $\rightarrow$ use $()$)
5. Write the answer in interval notation

Examples:

Find the domain and the range of the following functions:

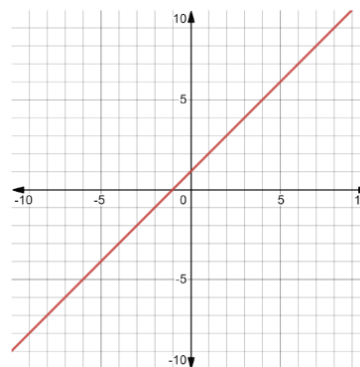
(a)



$$\text{Domain} = [-3, 3] \text{ or } -3 \leq x \leq 3$$

$$\text{Range} = [-3, 3] \quad -3 \leq y \leq 3$$

(b)



$$\text{Domain} = (-\infty, \infty) \text{ or } -\infty < x < \infty$$

$$\text{Range} = (-\infty, \infty) \text{ or } -\infty < y < \infty$$

Note: In MATH141 you will also represent domain and range using set notation for example

(i) $\{x : -3 \leq x \leq 3\}$ and (ii) $\{x : x \in R\}$

Set Notation (If you are interested)

A set is a collection of objects. The objects belonging to the set are called the **elements** or **members** of the set.

Sets are usually denoted with upper case letters, $A, B, X, Y, \dots$ while lower case letters are used to denote elements $a, b, x, y, \dots$ of a set.

Membership in a set is denoted as follows:

An element, a , belonging to the set A is written as $a \in A$ and if a , does not belonging to the set A we write $a \notin A$.

A set may be specified in essentially two ways:

1. The elements of the set are listed within braces, $\{ \}$, and separated by commas.

Example:

$A = \{1,3,5,7,9\}$ [The set A is the finite collection of odd integers, 1 to 9 inclusive]

$Z = \{1,2,3,\dots\}$

2. Sets may also be described in words.

Example:

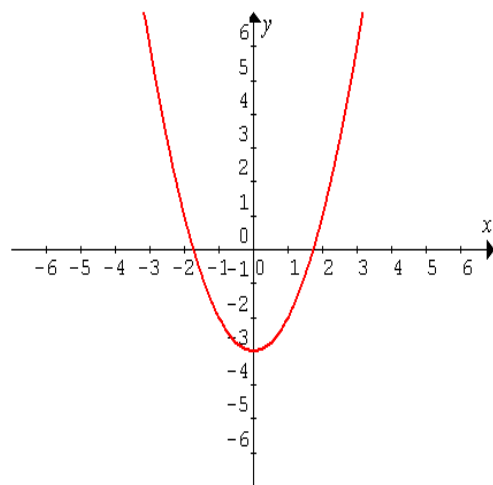
“the set of all real numbers”.

Alternately we can use the notation $\{x: \dots\}$ which is read “the set of all x such that”. Here the dots are replaced by the property that we wish x to have for this set. For the above example it would be written as $\{x : x \text{ is a real number}\}$.

Exercises:

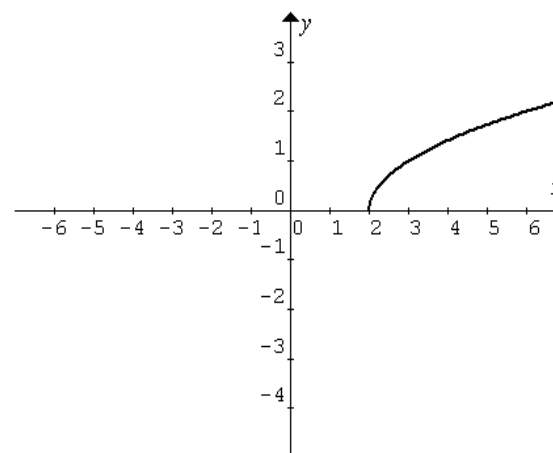
Find the domain and the range of the following functions:

(a) $y = x^2 - 3$



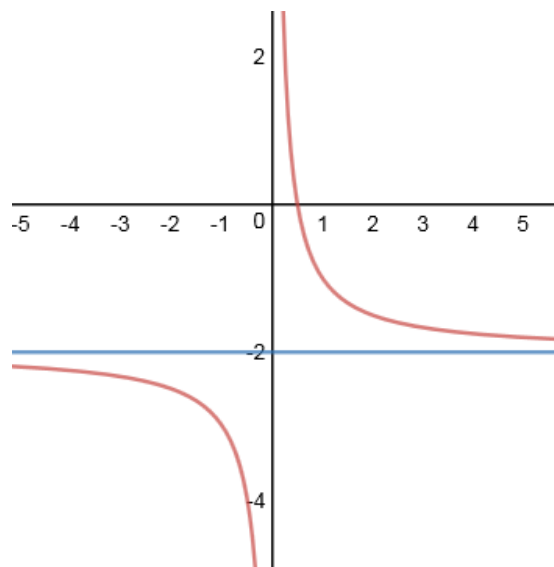
Domain: $(-\infty, \infty)$ or all x
Range: $[-3, \infty)$ or $y \geq -3$

(b) $y = \sqrt{x-2}$



Domain: $[2, \infty)$ or $x \geq 2$
Range: $[0, \infty)$ or $y \geq 0$

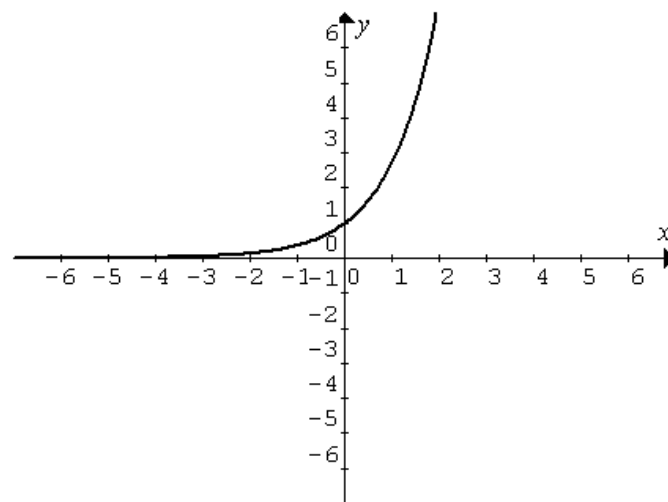
(c) $y = \frac{1}{x} - 2$



Domain: $(-\infty, 0) \cup (0, \infty)$
or
 $x \neq 0$

Range: $(-\infty, -2) \cup (-2, \infty)$
or
 $y \neq -2$

(d) $y = 2^x$



Domain: $(-\infty, \infty)$ or all x
Range: $(0, \infty)$ or $y > 0$

Key Idea – Finding Domain Algebraically

The domain is all x -values that can be put into the function.

Steps:

1. **Check for restrictions**, you will need to look for anything in the function that restricts which x -values can be used, for example:

Situation	Restriction	What to do
Straight Line	None	Domain is all values of x
Fraction	The denominator of a fraction cannot equal zero	Set the denominator $\neq 0$
Square Root	The expression under a square root sign cannot equal a negative number	Set the inside of the square root ≥ 0

2. Solve the inequality/equation to find forbidden x -values.
3. Write the domain in interval notation or set notation

Example: Find the domain of the following functions:

(a) $y = \frac{3}{x-2}$

The denominator of a fraction cannot equal zero

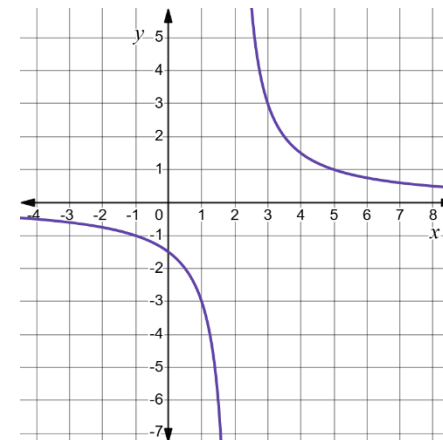
$$x - 2 \neq 0$$

$$x \neq 2$$

Domain :

Interval notation $(-\infty, 2) \text{ or } (2, \infty)$

Set notation $\{x : x \in R, x \neq 2\}$



(b) $y = \sqrt{x-4}$

The expression under a square root sign cannot equal a negative number

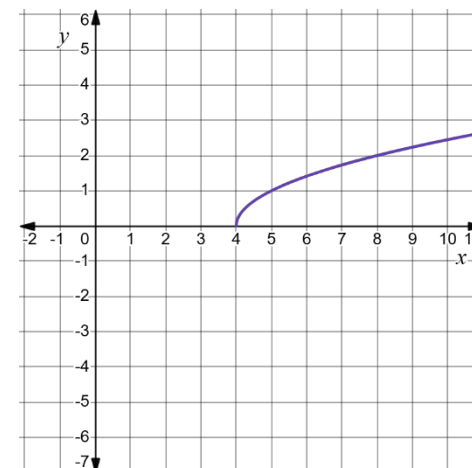
$$x - 4 \geq 0$$

$$x \geq 4$$

Domain :

Interval notation $[4, \infty)$

Set notation $\{x : x \geq 4\}$



$$(c) \ y = \frac{1}{\sqrt{4x+2}}$$

The expression under a square root sign cannot equal a negative number and the denominator cannot be zero

$$4x + 2 > 0$$

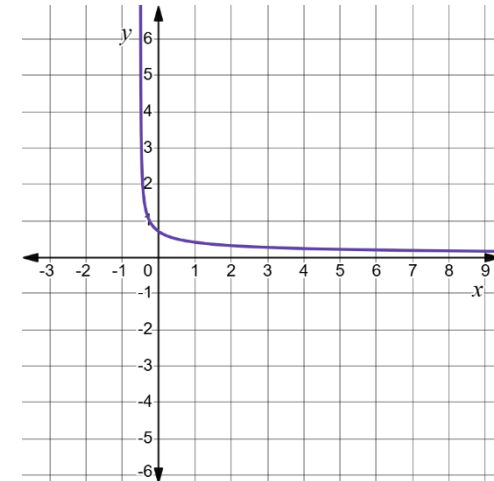
$$4x > -2$$

$$x > -\frac{1}{2}$$

Domain :

$$\text{Interval notation } \left(-\frac{1}{2}, \infty\right)$$

$$\text{Set notation } \left\{x : x > -\frac{1}{2}\right\}$$



Exercise:

Find the domain of the following functions:

(a) $y = -5x - 1$

Domain: $(-\infty, \infty)$ or all x

(b) $y = \frac{1}{2x+1}$

$$2x+1 \neq 0$$

$$2x \neq -1$$

$$x \neq -\frac{1}{2}$$

Domain: $(-\infty, -\frac{1}{2})$ or $(-\frac{1}{2}, \infty)$

(c) $y = \sqrt{2x - 4}$

$$2x - 4 \geq 0$$

$$2x \geq 4$$

$$x \geq 2$$

Domain: $[2, \infty)$

(d) $y = \frac{1}{\sqrt{x - 3}}$

$$x - 3 > 0$$

$$x > 3$$

Domain: $(3, \infty)$

(e) $y = \sqrt{2+x^2}$ (challenge question)

$$2+x^2 \geq 0$$

$$x^2 \geq -2$$

this is true for all x

Domain: $(-\infty, \infty)$

Key Idea – Finding Range Algebraically

The range is all y -values that are produced from the input values.

Steps:

1. **Check for restrictions**, you will need to look for anything in the function that restricts which y -values can be produced, for example:

Situation	Restriction	What to do
Straight Line	None	Range is all values of y
Fraction	The output can never be zero	Range is $y \neq 0$
Square Root	A square root cannot be negative	Range is $y \geq 0$

2. Write the range in interval notation or set notation

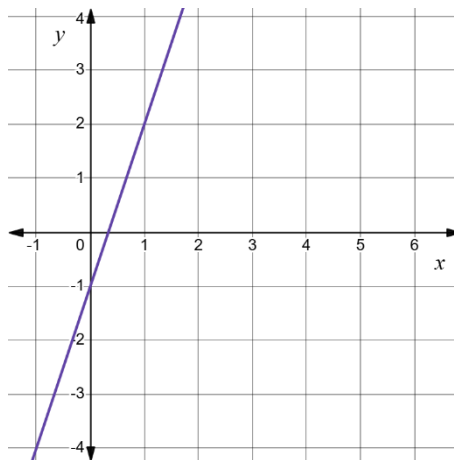
Example: Find the range of the following functions:

(a) $f(x) = 3x - 1$

Range:

Interval notation $(-\infty, \infty)$

Set notation $\{y : \text{all } y\}$

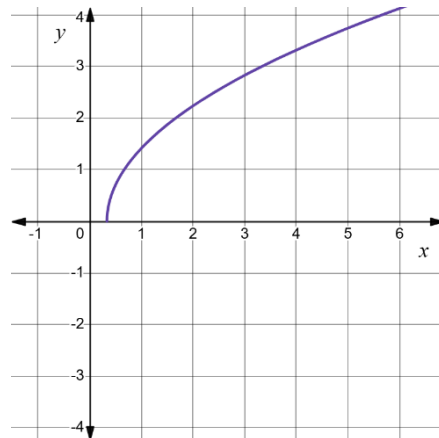


(b) $y = \sqrt{3x - 1}$

Range:

Interval notation $[0, \infty)$

Set notation $\{y : y \geq 0\}$

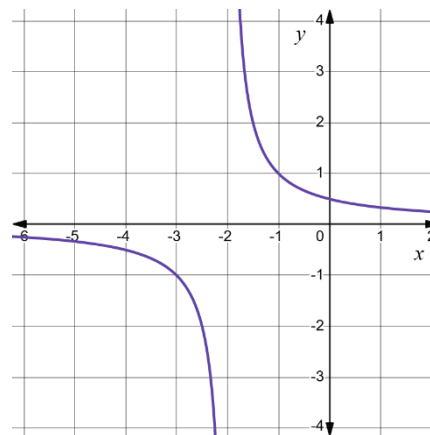


(c) $y = \frac{1}{x+2}$

Range:

Interval notation $(-\infty, 0) \text{ or } (0, \infty)$

Set notation $\{y : y \in \mathbb{R}, y \neq 0\}$



Exercise:

Find the range of the following functions

(a) $y = -5x - 1$

Range: $(-\infty, \infty)$ or all y

(b) $y = \frac{1}{x-2}$ $y \neq 0$

Range: $(-\infty, 0)$ or $(0, \infty)$

(c) $y = \sqrt{2x-4}$

$$y \geq 0$$

Range: $[0, \infty)$

(d) $y = \frac{1}{\sqrt{x-3}}$

$$y > 0$$

Range: $(0, \infty)$

Key Idea – Piecemeal Functions

A **piecemeal function** is made up of 2 or more formulae. The value of the piecemeal function depends on which formula is used, and this is determined by the values of x for which each formula is defined.

Example:

Given the piecemeal function

$$f(x) = \begin{cases} x^2 & \text{when } x \geq 0 \\ 1 - x & \text{when } x < 0 \end{cases}$$

From the function we can see that:

$$f(0) = 0^2 = 0, f(2) = 2^2 = 4 \text{ and } f(-4) = 1 - (-4) = 5$$

Exercise:

Given the piecemeal function:

$$f(x) = \begin{cases} x^2 - 4 & \text{when } x \geq 2 \\ -2x + 4 & \text{when } -1 < x < 2 \\ 6 & \text{when } x \leq -1 \end{cases}$$

Find $f(0) + 0.5f(3) - f(-2)$.

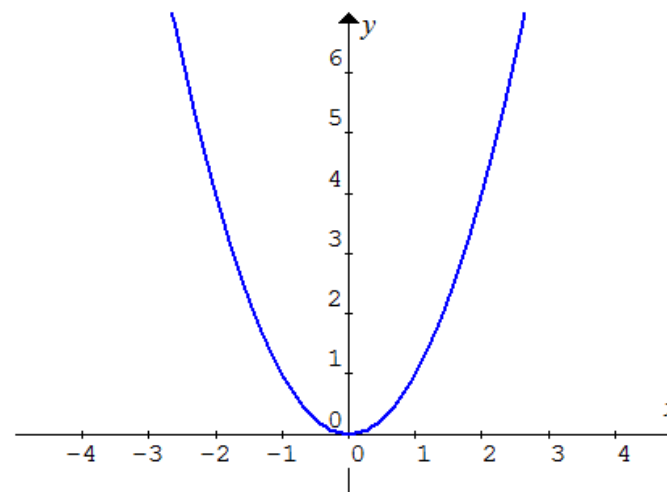
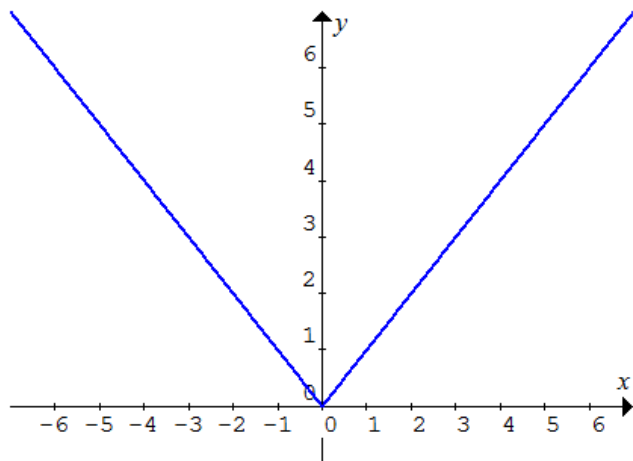
- $f(0) = -2 \times 0 + 4 = 4$
- $f(3) = 3^2 - 4 = 5$
- $f(-2) = 6$
- $f(0) + 0.5f(3) - f(-2) = 4 + 0.5 \times 5 - 6 = 0.5$

Key Idea – Odd and Even Functions

An **even function** is **symmetrical about the y axis**, i.e. where the graph of the right-hand part is the mirror image of that on the left. Even functions have the following property for all values of x :

$$f(-x) = f(x)$$

Example: The graphs of $y = |x|$ and $y = x^2$ are examples of even functions.



Example:

Show algebraically that $f(x) = x^4 + 3$ is an even function.

Need to show that $f(-x) = f(x)$

$$\begin{aligned} f(-x) &= (-x)^4 + 3 \\ &= x^4 + 3 \\ &= f(x) \end{aligned}$$

Hence $f(-x) = f(x)$ and so the function is even.

Example:

Show that $f(-3) = f(3)$ given the function $f(-x) = x^4 + 3$.

$$\begin{aligned} f(-3) &= (-3)^4 + 3 \\ &= 81 + 3 \\ &= 84 \\ &= f(3) \end{aligned}$$

Exercise: Show algebraically $f(x) = x^2 - 4$ is an even function.

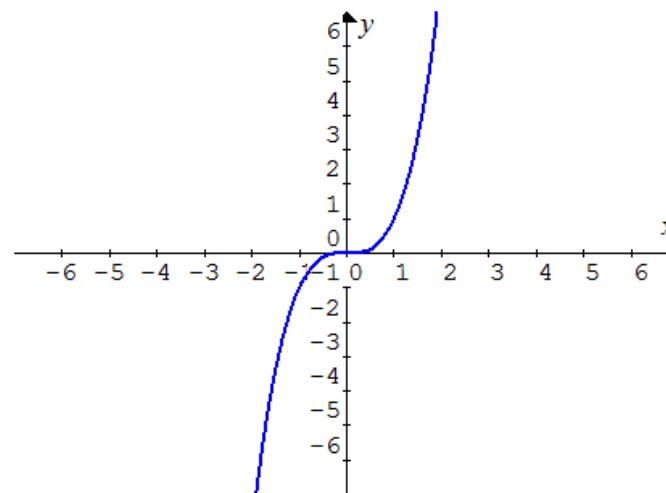
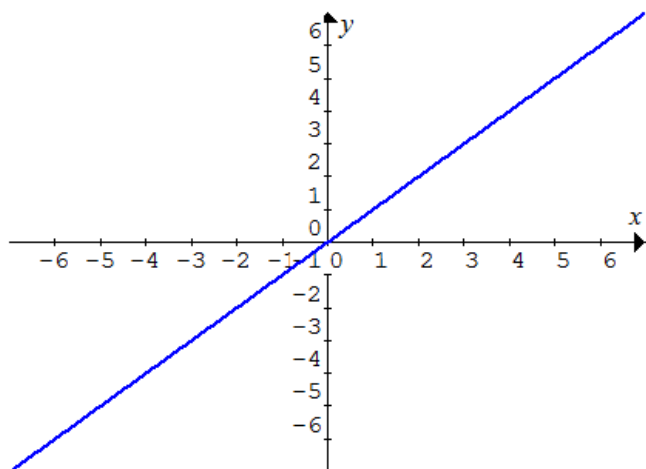
$$\begin{aligned} f(-x) &= (-x)^2 - 4 \\ &= x^2 - 4 \\ &= f(x) \end{aligned}$$

since $f(-x) = f(x)$ it is an even function

An odd function has a rotational symmetry about the origin, i.e. when rotated **180°** transforms into itself. Odd functions have the following property for all values of x :

$$f(-x) = -f(x)$$

Example: The graphs of $y = x$ and $y = x^3$ are examples of odd functions.



Example:

Show algebraically that $f(x) = x^3 + 3x$ is an odd function.

Need to show that $f(-x) = -f(x)$

$$\begin{aligned}f(-x) &= (-x)^3 + 3(-x) \\&= -x^3 - 3x \\&= -(x^3 + 3x) \\&= -f(x)\end{aligned}$$

Hence $f(-x) = -f(x)$ and so the function is odd.

Example:

Show that $f(-2) = -f(2)$ given the function $f(x) = x^3 + 3x$.

$$\begin{aligned}f(-2) &= (-2)^3 + 3(-2) & -f(2) &= -((2)^3 + 3(2)) \\&= -8 - 6 & &= -(8 + 6) \\&= -14 & &= -14\end{aligned}$$

Exercise: Show algebraically that $f(x) = x^5 - 2x^3$ is an odd function:

$$\begin{aligned} f(-x) &= (-x)^5 - 2(-x)^3 \\ &= -x^5 - 2 \times (-x^3) \\ &= -x^5 + 2x^3 \\ &= -(x^5 - 2x^3) \\ &= -f(x) \end{aligned}$$

since $f(-x) = -f(x)$ it's an odd function